

PRUDENCE MORAA NYANGWESO

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PROFESSIONAL SUMMARY

Detail-oriented and proactive Computer Science professional specializing in Data Validation, Quality Assurance (QA), and AI training data optimization. Proven ability to transform raw, chaotic datasets into highly reliable, structured pipelines for Machine Learning models. Deeply comfortable auditing complex datasets for classification anomalies, managing shifting regulatory data frameworks, and utilizing specialized collection infrastructure to ensure 100% data integrity.

CORE COMPETENCIES

- **Data Validation & Integrity:** Implementing rigorous checks and audits to catch structural errors, labeling flaws, and classification anomalies.
- **Dataset Auditing & Cleaning:** Processing, coding, and parsing complex, unstructured source datasets to train machine learning models using Python and SQL.
- **Mobile & Web Collection Logic:** Practical familiarity with managing production workflows, web surveys, and mobile collection tools (KoboToolbox, Google Forms).
- **Business Analysis**
- **Problem Solving & Strategic Decision Making**
- **Teamwork & Collaboration**
- **Process Improvement**
- **Technology & Digital Transformation**
- **Logic & Edge-Case Tagging:** Acute attention to detail developed through analyzing complex system states and asset classification pipelines

TECHNICAL SKILLS

- **Languages:** Python (Pandas, NumPy, Data Analysis Libraries), SQL, C#
- **Tools & Frameworks:** Microsoft Fabric, Flask, Git/GitHub
- **Databases & Cloud:** MySQL, PostgreSQL, SQL Server, AWS (Cloud Fundamentals), Render

PROFESSIONAL WORK EXPERIENCE

SAFRA DATA SCHOOL | *Data Quality & Business Intelligence Intern*

May 2025 – August 2025

- **Data Auditing:** Audited, processed, and validated raw, unstructured operational datasets, identifying and fixing structural anomalies to achieve 100% data accuracy for modeling teams.
- **Error Debugging:** Identified, troubleshooted, and independently patched high-priority data quality bottlenecks, formulating programmatic corrective constraints to maximize downstream dataset reliability.

- **Schema Mapping:** Authored technical documentation outlining validation protocols, ingestion schemas, and rigorous categorization rules to streamline validation pipeline handoffs.

TECHNICAL AI & LOGIC PROJECTS

AI-Powered Healthcare Insurance Risk Prediction System | *Developer & Data Investigator*

- **Feature Engineering & Labeling:** Structured data ingestion workflows, balanced sparse features, and managed cross-validation training loops to build a Logistic Regression model scoring 93% classification accuracy.
- **Dynamic Schema Adaptation:** Swiftly updated backend data structures mid-cycle to incorporate complex, shifting national regulatory rules (SHIF/SHA), ensuring compliant training labels.

Operational Systems & Logic Simulations | *Independent Creator*

- **Asset Pipeline Optimization:** Developed interactive logic trees, complex spatial positioning variables, and strict metadata rules for objects in multi-threaded runtime engines (Unity/Godot).
- **Rapid Edge-Case Tracking:** Refined sharp pattern recognition and bug-isolation techniques via extensive debugging cycles during dense framework compilation pipelines.

EDUCATION

UNIVERSITY OF EMBU — *Bachelor of Science in Computer Science (2022-2026)*

Graduated with second class honours (upper division)

RIAMBASE MIXED SECONDARY SCHOOL — *Kenya Certificate of Secondary Education (KCSE) (2018 – 2021)*

CERTIFICATIONS

- AWS AI Practitioner Challenge Completion (2026)
- Generative AI Foundation Certification (Udacity)
- IBM SkillsBuild: Data Analysis & Cloud Computing Fundamentals
- Business Intelligence and Data Analysis – Intermediate (Safra Data School)
- Microsoft Certified: Fabric Analytics Engineer Associate (DP-600)
- Microsoft Certified: Fabric Data Engineer Associate (DP-700) in progress
- Cyber Shujaa program, Data and AI track (in progress)

PROFESSIONAL REFEREES

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